

1. Palindrome Index

Given a string of lowercase letters in the range `ascii[a-z]`, determine the index of a character that can be removed to make the string a [palindrome](#). There may be more than one solution, but any will do. If the word is already a palindrome or there is no solution, return `-1`. Otherwise, return the index of a character to remove.

Example

$s = \text{"bcbc"}$

Either remove `'b'` at index **0** or `'c'` at index **3**.

Function Description

Complete the `palindromeIndex` function in the editor below.

`palindromeIndex` has the following parameter(s):

- *string s*: a string to analyze

Returns

- *int*: the index of the character to remove or `-1`

Input Format

The first line contains an integer q , the number of queries.
Each of the next q lines contains a query string s .

Constraints

- $1 \leq q \leq 20$
- $1 \leq \text{length of } s \leq 10^5 + 5$
- All characters are in the range `ascii[a-z]`.

Sample Input

STDIN	Function
3	$q = 3$
aaab	$s = \text{'aaab'}$ (first query)
baa	$s = \text{'baa'}$ (second query)
aaa	$s = \text{'aaa'}$ (third query)

Sample Output

```
3
0
-1
```

Explanation

Query 1: "aaab"

Removing `'b'` at index **3** results in a palindrome, so return **3**.

Query 2: "baa"

Removing `'b'` at index **0** results in a palindrome, so return **0**.

Query 3: "aaa"

This string is already a palindrome, so return **−1**. Removing any one of the characters would result in a palindrome, but this test comes first.

Note: The custom checker logic for this challenge is available [here](#).